PRISM v3

Pipeline Architecture and Model Documentation

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Abstract

Technical reference for the PRISM v3 codebase. Every section maps to a code cell; every hyperparameter is listed with its default. This document is the "how"; for the "why," see *PRISM: Mathematical Foundations*.

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1 Pipeline at a Glance

PRISM v3 lives in a single file (`prism_v3.py`), fifteen cells running end-to-end from raw Yahoo Finance data to a fully diagnosed backtest. The flow is linear:

Cell	What it does	Cell	What it does
1	Install deps	9	Training loop
2	Imports & logging	10	Predict + error tracking
3	Config dataclass	11	Allocation + OT
4	Order book tracker	12	Backtest engine
5	Download & clean	13	Vol-fit + metrics
6	Weekly resample + GK	14	31 diagnostic plots
7	Feature engineering	15	Summary dump
8	Stacked ensemble		

Everything downstream of cell 3 is parameterised by a single `@dataclass Config` (Section 2). No magic numbers live outside it.

2 Configuration

Parameter	Default	Role
Date boundaries		
`START_DATE`	2003-01-01	Download start
`END_DATE`	2025-12-31	Download end
`TRAIN_END`	2011-12-31	Last in-sample date
`TEST_START`	2012-01-01	First OOS date
Feature construction		
`WEEKLY_FREQ`	W-FRI	Resample anchor
`CORR_WINDOW`	26	Rolling correlation window (weeks)
`RETURN_LAGS`	[1,2,4,8,12]	AR lag set
`VOL_WINDOW`	8	Return-vol rolling window
`VOL_HISTORY`	26	Depth k of the V matrix
Model training		
`EWMA_DECAY`	0.94	Error-history decay
`TUNING_TRIALS`	10	Optuna trials per model
`TUNING_TIMEOUT`	20 s	Optuna wall-clock cap
`CV_SPLITS`	5	`TimeSeriesSplit` folds
`TORCH_EPOCHS`	100	MLP max epochs
`TORCH_PATIENCE`	12	MLP early-stop patience
`TORCH_BATCH`	64	MLP batch size
Allocation		
`LONG_ONLY`	False	If true, $w_i \geq 0$

Parameter	Default	Role
W_MAX	0.20	Single-asset cap
L_MAX	1.6	Gross leverage cap
SIGMA_TARGET	0.15	Ann. vol target
RF_ANNUAL	0.045	Risk-free rate
Execution		
DEFAULT_TC_BPS	10	Transaction cost (bps)
SINKHORN_EPS	0.05	OT regularisation scale
SINKHORN_ITER	200	Max Sinkhorn iterations
SINKHORN_TOL	10^{-8}	Convergence threshold
VOL_LOOKBACK	26	Vol-target lookback
VOL_CAP	2.5	Max vol-target leverage

3 Asset Universe

24 tickers covering global equity indices and commodities. Any ticker with fewer than 100 daily observations after cleaning is dropped, so n is determined at runtime.

Class	Region	Tickers
Equity	N. America	^GSPC, ^NDX, ^DJI, ^RUT, ^GSPTSE
	Europe	^FTSE, ^GDAXI, ^STOXX50E, ^FCHI, ^IBEX
	Asia-Pac	^N225, ^HSI, ^AXJO, ^BSESN, ^NSEI, ^KS11, ^TWII
	LatAm	^BVSP, ^MXX
Vol	N. America	^VIX
Commodity	Global	GC=F, SI=F, CL=F, NG=F

4 Data Pipeline

4.1 Download and cleaning (Cell 5)

Raw daily OHLCV comes from Yahoo Finance (`yfinance`, `auto_adjust=False`). Cleaning is deliberately conservative:

1. Forward-fill Close. Backfill Open/High/Low from Close if missing. Volume defaults to 0.
2. Drop any row that still has no Close.
3. Discard any ticker with ≤ 100 surviving daily bars.

4.2 Weekly resampling (Cell 6)

Aggregation is anchored on Friday close: O = first, H = max, L = min, C = last, V = sum. Weekly returns and Garman–Klass volatility are computed per the definitions in *Math Foundations*, Section 1.

4.3 SPY benchmark

SPY is downloaded separately (`auto_adjust=True`), resampled to the same weekly grid, and aligned to the test-period index. It is never part of the tradable universe.

5 Feature Engineering (Cell 7)

Two feature sets are built, one for each prediction task.

Pair features (10-dim)

One row per unordered pair (i, j) per week. Three lags of rolling correlation, both assets' returns and GK vols, a vol ratio, and a cross-vol term. Target: next-week correlation $\hat{\rho}_{ij,t+1}$.

Return features (14-dim)

One row per ticker per week. Five lagged returns, three rolling means (4/12/26 w), two rolling vols (8/12 w), GK vol, vol-of-vol, and return acceleration. Target: next-week return $r_{i,t+1}$.

Both feature sets are described equation-by-equation in *Math Foundations*, Section 3.

`[Insert feature dimension summary from execution output.]`

6 Ensemble Architecture (Cell 8)

6.1 Base learners

Nine models, loosely ordered from simple to complex:

#	Model	Tuned?	Key defaults
1	OLS	No	—
2	Ridge	Yes	$\alpha = 1.0$; search $[10^{-4}, 100]$
3	Lasso	Yes	$\alpha = 0.001$, `max_iter`=5000
4	ElasticNet	Yes	$\alpha = 0.001$, ℓ_1-ratio= 0.5
5	Random Forest	Yes	250 trees, depth 6, min-leaf 5
6	Extra Trees	Yes	250 trees, depth 8, min-leaf 3
7	GBM	Yes	300 rounds, depth 3, $\eta = 0.05$
8	AdaBoost	No	100 estimators, $\eta = 0.05$
9	MLP	No	3×64 hidden, ReLU, dropout 0.1

Tuning uses Optuna (TPE sampler, seed 42) with 10 trials or 20 s wall-clock, whichever binds first. CV is `TimeSeriesSplit(5)`, scored by MSE.

6.2 MLP details

Listing 1: MLP architecture (identical for pair and return tasks).

```
layers = []
in_dim = d   # 10 (pair) or 14 (return)
for _ in range(3):
    layers += [Linear(in_dim, 64), ReLU(), Dropout(0.1)]
    in_dim = 64
layers.append(Linear(64, 1))
# Adam, lr=1e-3, weight_decay=1e-4, batch=64, patience=12
```

6.3 Stacking

After the base learners are trained on the first 80% of in-sample data, they predict on the held-out 20% to produce a $2M$-dimensional meta-feature vector: each base model's prediction, plus its absolute error. A Ridge meta-model ($\alpha = 1.0$) learns the final combination.

At test time, the absolute-error half is zeroed out (the true target is unavailable), so the meta-model falls back to a learned linear blend of the raw predictions.

6.4 Training procedure

For each pair or ticker:
1. Filter to training period. Require ≥ 50 rows.
2. 80/20 chronological split into base and meta sets.
3. `StandardScaler` fit on the base set, applied to both.
4. Train all 9 base learners (with Optuna where marked).
5. Predict on the meta set $\rightarrow$ meta-features.
6. Train Ridge meta-model.
7. Store scaler + all models.

Per-model diagnostics (MSE, MAE, R^2, directional accuracy) are written to `MODEL_DIAG` for later plotting.

7 Prediction and Error Tracking (Cell 10)

At each test date t, the trained ensembles produce:
- $\hat{\mu}_{t+1} \in \mathbb{R}^n$ (return predictions, unclipped).
- $\hat{C}_{t+1} \in \mathbb{R}^{n \times n}$ (correlation predictions, clipped to $[-1, 1]$).

Error histories grow as lists per ticker/pair. At each step we append the signed error $\varepsilon =$ predicted $-$ actual and recompute the five error statistics (EWMA bias, std, skewness, kurtosis, directional hit) that feed the confidence module. The EWMA uses $\lambda = 0.94$; higher moments come from `scipy.stats` on the full history.

8 Allocation Engine (Cell 11)

Five stages run in sequence for each test week. The math is in *Math Foundations*, Sections 4–9; here we document the implementation.

Stage 1: Confidence (`compute_conf`). Reads E_μ, returns $\kappa \in \mathbb{R}^n$.
Stage 2: Bayesian posterior (`bayes_post`). NIW update $\rightarrow (\mu^B, \Sigma^B)$.
Stage 3: Topological analysis (`topo_analysis`). Persistent homology on the correlation distance graph, spectral denoising $\rightarrow C^{\text{spec}}$.
Stage 4: Robust covariance (`robust_cov`). EWMA vol from V, regime adjustment, blend with $\Sigma^B \rightarrow \Sigma_{\text{final}}$.
Stage 5: Sharpe allocation (`sharpe_alloc`). 50-point γ-sweep, CVXPY + SCS (`max_iters`=5000), pick γ^* by Sharpe $\rightarrow \boldsymbol{w}^*$.

Nearest-PD projection. Called after every covariance construction: eigendecompose, clamp eigenvalues $\geq 10^{-8}$, reconstruct and symmetrise. If the minimum eigenvalue is still below 10^{-8}, add a diagonal ridge.

Fallback. If any stage throws, the allocator returns equal-weight ($\boldsymbol{w} = n^{-1}\mathbf{1}$) and logs a warning.

9 Order Book Tracker (Cell 4)

This is a simulation layer, not a live feed. It reconstructs synthetic L2 snapshots from weekly OHLCV so that the backtest can estimate realistic spread and slippage costs.

Snapshot construction. Mid = Friday close. Spread $\approx \max(\hat{\sigma}^{\mathrm{GK}} \times 0.1,\ 10^{-4})$. Five depth levels with geometrically decaying size ($1000 \times 0.8^{\ell}$, $\ell = 0, \ldots, 4$).

Rebalance logging. Each off-diagonal OT flow $T^*_{ij} > 10^{-8}$ is logged as a fill, recording: date, sell/buy assets, flow %, notional, spread, slippage. Weekly summaries (fill count, total notional, total slippage, turnover %) feed the diagnostic plots.

10 Backtesting Engine (Cell 12)

One iteration per test week:
1. Retrieve actual weekly returns.
2. Build order book snapshot.
3. Run the five-stage allocation pipeline.
4. Compute OT transport plan and transaction cost $\mathcal{C}^*$.
5. Log the rebalance.
6. Raw return: $r^{\mathrm{raw}} = (\boldsymbol{w}^*)^\top \boldsymbol{r} - \mathcal{C}^*$.
7. Vol-target scaling (lookback 26 w, cap 2.5×).
8. Record diagnostics: weights, κ, μ^B, σ, leverage, turnover.

11 Vol-Fitting and Metrics (Cell 13)

Three strategies — PRISM, equal-weight, and SPY — are all vol-fitted to 15% annualised using the same mechanism (26-week lookback, cap 3.0×). This gives a fair apples-to-apples comparison at identical risk budgets.

Metrics computed (per strategy): total return, annualised return, annualised vol, Sharpe, Sortino, max drawdown, Calmar, win rate, skewness, kurtosis.

12 Diagnostic Plots (Cell 14)

31 plots, grouped thematically.

	Plot	Shows
Performance		
P01	Cumulative returns	All strategies, log scale
P02	Vol-matched comparison	PRISM / EW / SPY at 15%
P03	Drawdown panels	Three strategies stacked
P04	Rolling Sharpe (26 w)	
P05	Rolling return (52 w)	
P06	Rolling vol + target	26 w realised vs. 15% line
Portfolio composition		
P07	Weight stacks	Long / short, stacked area
P08	Leverage	Gross and net
P09	Turnover + TC	Weekly bars
P22	Weight heatmap	Ticker $\times$ time
Return analysis		
P10	Return distributions	Histogram overlay
P11	QQ plots	vs. normal
P12	Monthly heatmap	Calendar grid
P23	Annual returns	Grouped bar chart
P24	Underwater plot	
Model diagnostics		
P13	κ evolution	10 representative tickers
P14	Avg κ + predicted vol	Dual-axis
P15	μ^B heatmap	
P16	σ heatmap	
P17	Final correlation + κ	Last test date snapshot
P18	Return R^2 per ticker	
P19	Return DA per ticker	
P20	Meta-model weights	Ridge coefficients
P21	Pair model diagnostics	R^2, DA, MAE
Market and execution		
P25	Average GK vol	Cross-sectional
P26	All assets (log)	Normalised to 1
P27	Individual asset panels	Small multiples
P28	Strategy vs. benchmarks	At 15% vol
P29	Order book summary	Fill counts, notional
P30	Spread history	
P31	Last trades	Net flows, final rebalance